AMRITA VISHWA VIDYAPEETHAM
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COIMBATORE CAMPUS

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

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Problem Statement

Secure communication systems today can encrypt and detect threats, but cannot autonomously recover a compromised session, and their RSA/ECC foundations are vulnerable to quantum attacks. We propose an **Agentic AI-driven, self-healing secure communication framework using Post-Quantum Cryptography (ML-KEM)** that detects compromised or abnormal sessions and autonomously restores and verifies them, without manual intervention.

Problem Understanding

- **Who:** Organisations relying on long-lived confidential communication (banking, healthcare, government, enterprise) and the security teams who must respond manually to every compromised session.
- **Cause:** Dependence on RSA/ECC, breakable by quantum computers (Shor's algorithm), and on alert-only AI that detects threats but cannot act; recovery is left to humans.
- **Impact:** Longer downtime and data exposure, encrypted data harvested today and decrypted later (*Harvest Now, Decrypt Later*), and overloaded security teams.

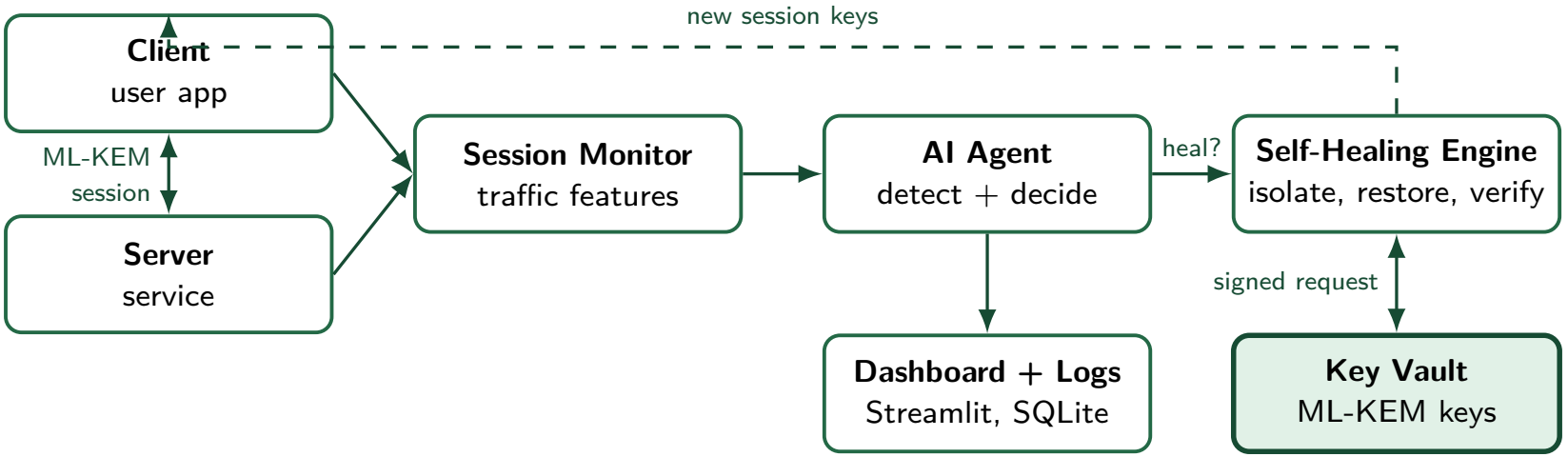
Existing Gaps & Challenges

- **Detection without autonomous action:** AI-based intrusion detection raises alerts, but isolating, re-keying and restoring sessions still need human intervention.
- **Prevention and recovery in silos:** PQC research focuses on quantum-safe keys; self-healing research ignores quantum threats. Recovered sessions are not re-secured with quantum-safe keys.
- **Unsolved challenge, trusting the agent:** Agentic AI can be fooled by adversarial attacks and acts as a black box. It must heal autonomously without becoming the weakest link or a single point of failure.

Our Proposed Solution

- **Core idea:** An agentic AI security agent that continuously monitors ML-KEM-secured sessions and autonomously heals any compromised or abnormal session.
- **How it works:** A closed loop, **Observe** $\rightarrow$ **Detect** $\rightarrow$ **Isolate** $\rightarrow$ **Re-key** $\rightarrow$ **Restore** $\rightarrow$ **Verify**. Fresh ML-KEM keys are generated in an isolated HSM-style key vault, so the agent never holds session keys.
- **Value:** Recovery in seconds instead of hours, quantum-safe protection, reduced load on security teams, and a protected agent with limited permissions, rate limits and audit logs.

Technical Approach



The monitor feeds session behaviour to the AI agent; on an anomaly the agent asks the self-healing engine to isolate the session, the key vault issues fresh ML-KEM keys, and the session is restored and verified. The agent never sees the keys.

Tech Stack

Frontend: Streamlit dashboard

Backend: Python,liboqs-python (ML-KEM)

Database: SQLite (session + action logs)

AI/ML: Scikit-learn, CICIDS2017 dataset

Innovation & Uniqueness

- **What is new:** Existing work treats PQC (prevention) and AI self-healing (recovery) separately. We combine them, so every healed session is **re-secured with fresh quantum-safe ML-KEM keys**.
- **Why it suits the problem:** An agentic AI does not stop at an alert; it observes, decides, acts and verifies on its own, removing the slow manual step that leaves sessions exposed.
- **Key differentiator, a protected agent:** The agent's own key is locked in an HSM-style vault and it can only send signed re-key requests. It never holds session keys, has limited permissions and is rate-limited, so even a compromised agent cannot leak keys.

Feasibility & Scalability

- **Feasibility:** Built fully in software with free, open-source tools (Python, liboqs-python, Scikit-learn, Streamlit) on our own laptops. Steps: (1) ML-KEM secure session, (2) anomaly detection model, (3) self-healing loop, (4) key vault and dashboard.
- **Scalability:** One lightweight agent per server or gateway; agents can run as containers in the cloud, and a real hardware HSM/TPM can replace the prototype vault in deployment.
- **Risks and mitigations:** False alarms $\rightarrow$ verification step before healing; adversarial attacks on the AI $\rightarrow$ limited permissions, rate limits and audit logs; PQC overhead (larger keys) $\rightarrow$ ML-KEM is lattice-based and efficient.

Use Cases 1

Scenario 1: Interbank payment link

- **User :** Priya, a payments operations engineer at a global bank
- **Situation :** During a busy settlement window, a high-value transfer session to a partner bank starts showing odd timing and message patterns.
- **What happens:** The agent flags the session as anomalous, isolates it, and starts a fresh of ML-KEM key exchange over a clean channel. Verifies integrity of the messages, replays any unconfirmed transactions, and logs everything.
- **Outcome:** Priya gets an alert with the full audit trail after the session is already restored. Settlement continues with no manual intervention.

Use Cases 2

Scenario 2: Tactical field communications

- **User:** Sgt. Alvarez, a unit communications lead
- **Situation:** A drone feed and command link are being jammed, and a rogue node is trying to join the mesh network.
- **What happens:** The agent detects the spoofed node, quarantines it, and revokes its session keys. It reroutes traffic through an alternate relay, renegotiates PQC-protected keys, and verifies the endpoints before resuming.
- **Outcome:** Alvarez sees the link drop for seconds, not minutes, and the intercepted traffic can't be decrypted later by a quantum-capable adversary.

Use Cases 3

Scenario 3: Remote patient monitoring

- **User:** Dr. Chen, a cardiologist monitoring patients with pacemakers remotely
- **Situation:** A patient's pacemaker, which will be in service for 10+ years, begins sending telemetry through a session showing signs of tampering.
- **What happens:** The agent pauses the suspicious session, falls back to a safe read-only mode on the device, and re-authenticates it with fresh PQC credentials. It verifies the device's firmware and data integrity, then resumes.
- **Outcome:** Dr. Chen is notified of a verified recovery, with no gap in critical monitoring and the patient's data is kept private.

Impacts and Benefits

Benefits

- **Quantum-safe:** ML-KEM + AES-256 blocks "harvest now, decrypt later"
- **Protected agent:** PQC stops anyone tampering with the AI
- **Automated Detection:** Telemetry and AI spot anomalies instantly without manual oversight.
- **Self-Healing:** Automatically rekeys, re-authenticates, and reroutes in seconds.

Impacts

- Data stays safe long-term, even against future quantum attacks
- Security that runs on its own for machine-to-machine systems
- Quantum-resistant *and* able to recover from compromise

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG 9	Target 9.1	Industry, Innovation and Infrastructure	Builds reliable, resilient digital communication infrastructure
SDG 9	Target 9.5	Industry, Innovation and Infrastructure	Applies new research (PQC, agentic AI) to real systems
SDG 16	Target 16.6	Peace, Justice and Strong Institutions	Keeps institutions' data and services secure and trustworthy

Individual Member Contribution

Member	Contribution or responsibility
Avantika S	ML-KEM secure session and key vault (post-quantum cryptography). Testing, ideator, research and documentation
Lakshmi Krishna	AI anomaly detection model, training, re-key and testing, research and documentation
Sreeparvathy S Menon	Agentic AI, isolate, restore, verification
Sruthy Devaraj	Self healing methodology, testing, error analysis

Yukti ID

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